Examiner reports

Q1.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and ½ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution. Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of f(0.4) and f(0.5). Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when the tan became removed from its $2x$ and $\frac{1}{2}x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the sin $2x$ often got the method marks. There were candidates who started with u = x and v = cos $2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.